DMITRII VASILEV

Open systems. Measured results.

CONTACT

Ko Samui, Thailand (UTC+7)
admin@t27.ai
Telegram: @t27_dev
github.com/gHashTag

OPEN TOOL FLOW

- Verilog + Icarus
- Yosys + nextpnr-xilinx
- prjxray + openXC7
- openFPGALoader + JTAG/XVC
- Python golden models

SOFTWARE

Python, Rust, Zig, Tcl, Git, Linux/Unix, CI, compiler and code-generation workflows

WORKING MODE

Remote full-time or contract. English B2. Technical writing and correspondence in English.

HONEST GAPS

No production ECP5/Lattice, Kestrel/LiteX, or signal-integrity ownership is claimed. Current board work is Artix-7.

EDUCATION

SADK - Technician-technologist (higher), 1988.

State University of Management, Moscow - Management, 2006.

MID LEVEL FPGA ENGINEER - TARGETED APPLICATION

OPEN-SOURCE FPGA AND VERIFICATION ENGINEER

10+ years in production software, now focused on FPGA/RTL and verification. Recent work spans protocol-driven HDL, bit-exact reference models, synthesis and place-and-route debugging, board bring-up and formal proofs. Negative results and hardware regressions are recorded, not hidden.

CURRENT-TREE EVIDENCE

29 MERGED OPENXC7 PRs | 27 IN NEXTPNR | 26 STILL LIVE

- One nextpnr change was reverted after a hardware regression. Both the merged and current-tree counts are disclosed.
- Built an end-to-end Artix-7 flow with Verilog, Python golden models, Icarus, Yosys, nextpnr-xilinx, prjxray, openFPGALoader and JTAG.
- Work includes timing closure, directed and known-answer vectors, bit-exact regression and reproducible bitstream builds.
- Reads protocol specifications and converts them into checked implementation boundaries: recent work includes an FPGA network stack and AD9361 radio path.

RECENT TECHNICAL WORK

FPGA / RTL / Verification Engineer - Trinity S3AI | 2026-present

- Designed and verified ternary neural primitives and a GF16 matrix-multiply design for Artix-7; every RTL node is compared against an independent Python model.
- Built a spec-first compiler path that emits bit-exact Verilog and software backends from one source of truth, with independent checks intended to catch specification/RTL divergence before synthesis.
- Measured a ternary network whose dot-product path is multiplier-free. Published boundary: post-synthesis, DSP inference disabled; no post-route timing or energy claim.
- Prepared a TinyTapeout SKY130 design through GDS generation, gate-level test and precheck. No fabricated-die claim is made.

AI Consultant / Agent-systems Developer | 2025-2026

- Delivered multi-agent software systems and mentored more than 50 developers, following earlier production work in mobile, payments, health technology and developer education.

VERIFICATION AND PUBLIC WORK

Verification: Python golden models, Icarus simulation, bit-exact conformance, directed and known-answer vectors, regression, CDC awareness, Coq/Rocq. The current algebraic proof artifact compiles without Admitted declarations.

Selected work: trinity | trinity-fpga | ternary-network-floats | arXiv:2606.05017, arXiv:2606.09686

WORKING APPROACH

- Specification to independent model to RTL simulation to synthesis/P&R; to board evidence.
- Current-tree truth beats a clean-looking merged total; regressions and reverts stay visible.
- Results are labeled at their actual stage: synthesis is not P&R;, and GDS/precheck is not production hardware.

AVAILABLE FOR INTERNATIONAL REMOTE FULL-TIME OR CONTRACT WORK FROM THAILAND (UTC+7)